**BUSINESS ANALYSIS PORTFOLIO CASE STUDY**

# CarePlus Hospital Network

*DIGITAL PATIENT INTAKE TRANSFORMATION & GAP ANALYSIS*

Project Pulse | Healthcare & HealthTech | Discovery, Process Improvement and Requirements

Prepared for professional portfolio presentation • October 2026

| CURRENT STATE | PROPOSED FUTURE STATE |
|---|---|
| Paper registration, four-page forms, manual re-entry into legacy system | Pre-arrival mobile intake or assisted on-site iPad registration connected securely to existing EHR |

*Case-study status: analytical proposal, not an implemented hospital system. Quantitative baseline figures are from the supplied 10Alytics case brief; targets and implementation plans are proposed by the analyst.*

## 1. Executive Summary

CarePlus Hospital Network, a regional provider operating for more than 20 years, has a manual patient registration process that undermines its otherwise strong clinical offering. In the supplied case, patients complete four-page paper forms, nurses re-enter details in an older computer system, and information moves through paper-based handoffs. Registration can take up to two hours, medical-record errors are reported at 15%, and nurses spend 40% of their shift typing forms. Patient dissatisfaction and staff burnout have intensified as competing urgent-care clinics provide more convenient digital experiences.

Project Pulse evaluates the current workflow, defines the desired digital intake experience, identifies people, process and technology gaps, and translates findings into recommendations and an initial requirements baseline. The recommended direction is secure mobile pre-registration and assisted iPad check-in, integrated with the existing Electronic Health Records (EHR) system, supported by staff training, accessibility measures and continuous performance tracking.

This portfolio report demonstrates business problem framing, As-Is/To-Be mapping, gap analysis, Fishbone, SWOT, McKinsey 7S, PESTLE, stakeholder identification, business and solution requirements, risk analysis, prioritization and phased implementation planning. The work represents a structured analytical recommendation, not evidence of implementation or realized savings.

## 2. Case Context and Scope

### 2.1 Business context

CarePlus has experienced doctors but faces patient attrition to faster urgent-care providers. Its front-desk bottleneck creates avoidable delays, duplicate administrative work and opportunities for inaccurate patient records. The discovery brief contains comments from the Chief Medical Officer, Head Nurse and IT Lead calling for remote mobile registration or iPad registration in the waiting area, connection to the existing EHR, support for older patients and compliance with HIPAA as specified in the scenario.

### 2.2 Project objective

Identify what prevents CarePlus from moving from paper intake to a safe, inclusive, integrated digital workflow, and recommend a feasible transformation path that improves registration efficiency, information quality and patient experience.

### 2.3 Scope boundaries

| In scope | Out of scope / not demonstrated |
|---|---|
| As-Is and To-Be intake process, gap and root-cause assessment | Build or deployment of production mobile app or EHR integration |
| Stakeholder needs, requirements, risks and conceptual roadmap | Confirmed financial ROI or validated savings |
| Security/privacy, accessibility and adoption considerations | Clinical diagnosis, prescribing or triage decision automation |
| Suggested performance KPIs and test scenarios | Verified actual improvements or completed user acceptance testing |

## 3. Problem Definition and Evidence

| Case brief evidence | Business impact / implication |
|---|---|
| Up to 2 hours to register | Delay in accessing care; lower patient satisfaction |
| 15% medical-record error rate | Potential information-quality and patient-safety concerns |
| Nurses spend 40% of shifts entering forms | Clinical staff diverted from patient care |
| Four-page paper forms and manual retyping | Duplicate work, illegibility and handoff delays |
| Untrained front-desk staff to troubleshoot frozen iPads | Digital adoption and service continuity risk |
| Older patients may be unfamiliar with iPads | Accessibility and assisted service requirement |

### Problem statement

CarePlus lacks a coordinated, digitally supported intake process. Manual completion, transcription and transfer of patient information create excessive registration time and records errors, intensify nurse workload and make the hospital less competitive. Any solution must retain access to historical EHR data and accommodate patients who need assistance.

*The source reports "up to two hours," not a verified mean. Do not treat this as an average unless additional data become available.*

## 4. Methodology and Analytical Approach

1. Review the project brief and discovery notes to identify observed pain points, constraints and stakeholder concerns.
2. Map the current registration workflow and articulate the desired future process.
3. Use the People–Process–Technology (PPT) model to define gaps and potential changes.
4. Use Fishbone, SWOT, McKinsey 7S and PESTLE to examine internal root causes, organizational alignment and external factors.
5. Translate findings into business goals, functional/non-functional requirements, priorities and validation criteria.
6. Recommend phased implementation, risk controls, pilot measures and next decisions for stakeholders.

*This is desk-based analysis of a supplied training case. No live staff interviews, process observation, EHR architecture inspection or independent hospital data audit were performed.*

## 5. Current-State Process Mapping (As-Is)

Patient arrives → queues at reception → receives four-page clipboard form → completes handwriting → submits form to front-desk nurse → nurse manually retypes into legacy hospital system → patient information is transferred to clinicians through siloed communication.

| Process step | Owner | Observed weakness |
|---|---|---|
| Queue and form collection | Patient / reception | Wait and paperwork burden |
| Handwritten information capture | Patient | Unreadable, incomplete or inconsistent entries |
| Manual data transcription | Front-desk nurse | Duplicate effort; transcription errors |
| Information access and handoff | Nurse / physician | Paper-based coordination and delays |

### As-Is diagnosis

The critical failure point is not the paper form alone. It is the dependence on repeated human handling of the same information with no verified automated data transfer between intake and EHR. The brief does not establish exactly where the two-hour delay accumulates; timing study and queue observation are required before allocating delay by step.

## 6. Proposed Future-State Process (To-Be)

Patient chooses mobile pre-registration or assisted on-site iPad → enters or verifies demographic and intake information → form checks required fields → patient submits securely → authorized integration validates and updates the matching EHR patient record → staff review exceptions and confirm check-in → physician accesses updated information according to role permissions.

### Key design principles

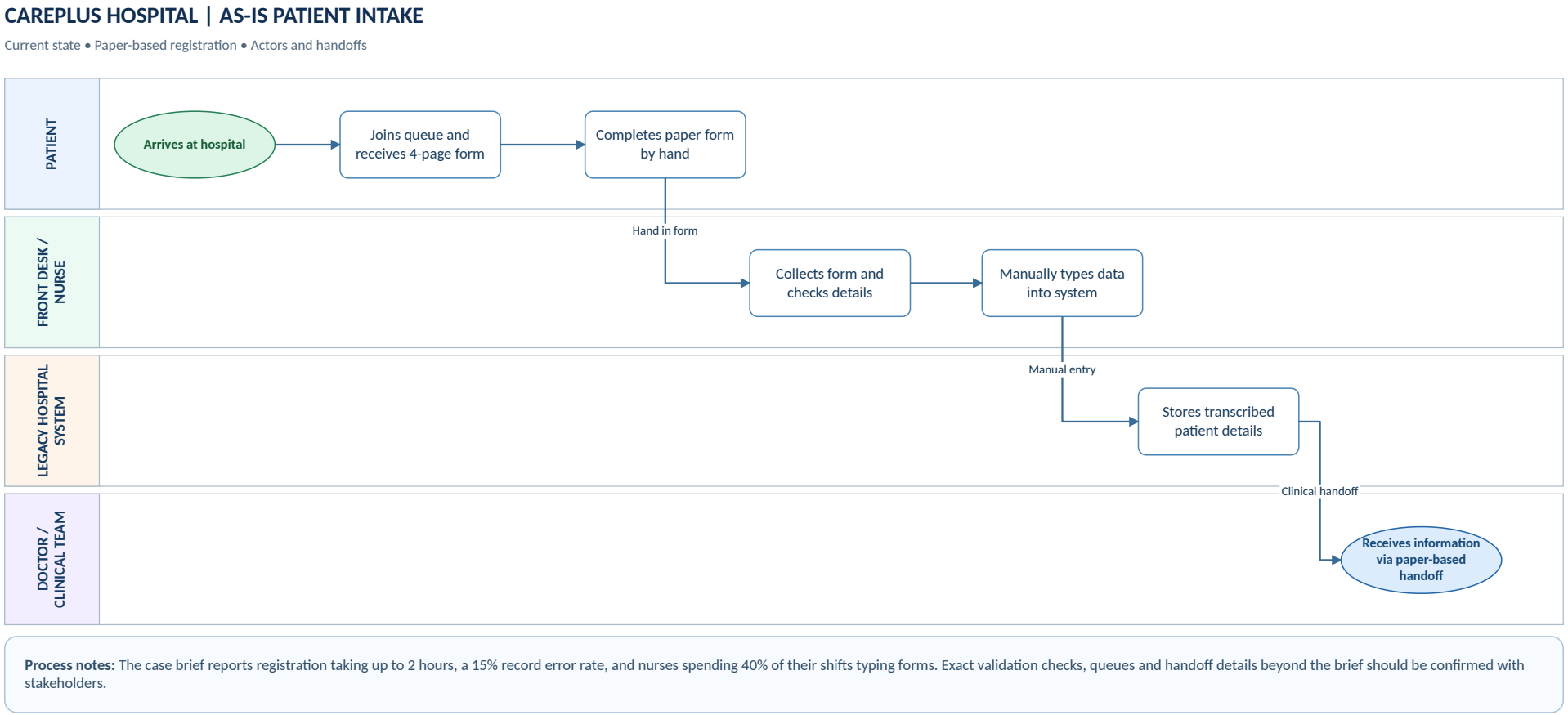
- One-time capture of patient information where possible, with duplicate-record checks.
- Assisted registration for older patients, people with disabilities or those without smartphones.
- Existing EHR history retained, with authenticated patient matching.
- Human review of errors and exceptions rather than silently overwriting records.
- Privacy and confidentiality built into the workflow, including consent and access control as applicable.
- Operational contingency for network, device or EHR downtime.

## 7. Gap Analysis

| Dimension | As-Is gap | Required To-Be capability |
|---|---|---|
| People | No front-desk device troubleshooting skills; patients may need assistance | Role-based training, digital guides and clear escalation support |
| Process | Handwritten forms, manual re-entry and siloed transfer | Redesigned check-in with single capture, validation and exception handling |
| Technology | No stated mobile/iPad intake link to legacy EHR | Supported integration, device management, secure data transfer and testing |
| Data and compliance | Transcription errors and sensitive | Data-quality validation, patient |

# Figure 1. As-Is Swimlane: Current Patient Intake

**SUPPORTS SECTION 5 | CURRENT-STATE PROCESS MAPPING (AS-IS)**



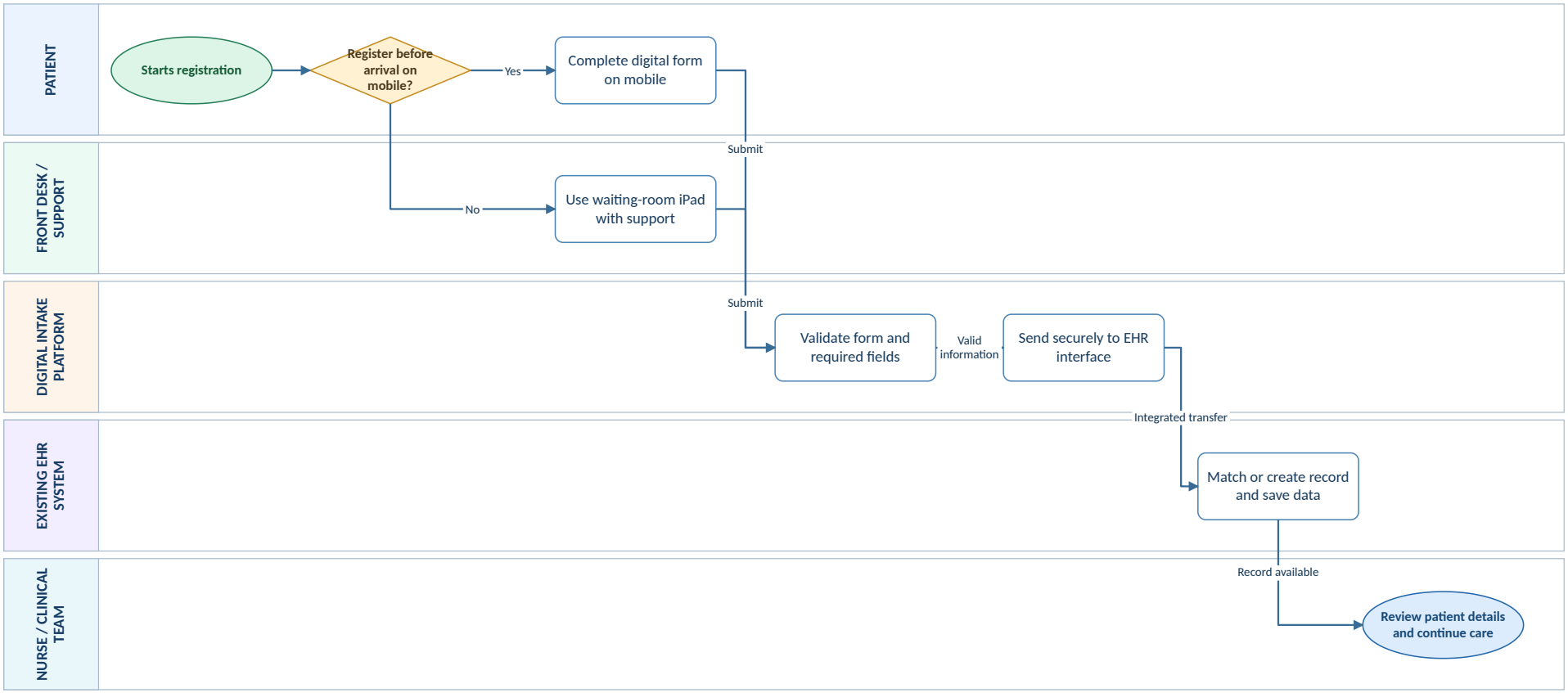

*Current state: paper registration, manual retyping and a paper handoff to clinicians. Figures (up to 2 hours, 15% record errors, 40% of nurse shifts) are from the supplied case brief.*

# Figure 2. To-Be Swimlane: Proposed Digital Patient Intake

**SUPPORTS SECTION 6 | PROPOSED FUTURE-STATE PROCESS (TO-BE)**

## CAREPLUS HOSPITAL | TO-BE DIGITAL PATIENT INTAKE

Proposed future state • Two registration channels • Secure EHR integration • Assisted access




**Process notes:** This is a proposed workflow, not a deployed system. If data is incomplete, the platform should prompt correction before transfer. Front-desk staff support patients unfamiliar with iPads. The EHR connection and privacy controls require technical validation.

*Proposed future state: two registration channels (mobile pre-registration or assisted waiting-room iPad), validation, secure transfer to the existing EHR. A proposed workflow, not a deployed system.*

| | | |
|---|---|---|
| | health information | matching, access control and audit logging |
| Measurement | Baseline wait-time distribution and error causes not supplied | Defined metrics, timestamps, feedback loop and reporting ownership |

Priority gap: secure and reliable connection between digital intake and the existing EHR. Without it, mobile or iPad forms could simply become a new front end to the same manual re-entry problem.

## 8. Fishbone Root-Cause Analysis

Effect being investigated: long registration delays, data errors, patient complaints and nursing administrative overload.

| Potential cause category | Hypotheses to validate |
|---|---|
| People | High manual workload; unfamiliarity with device troubleshooting; varied patient digital literacy |
| Process | Long paper form; duplicated typing; sequential handoffs; lack of standardized exception handling |
| Technology / equipment | Legacy system constraints; unavailable digital intake interface; tablet connectivity and reliability concerns |
| Management | Insufficient change readiness, training plans or dedicated workflow ownership |
| Environment | Busy reception and distressed or unwell patients; potential congestion at peak times |
| Measurement | Insufficient time stamps, error classification or patient-experience tracking in supplied data |

*The Fishbone entries are plausible contributing factors, not proven causes. Prioritize testing with time-motion observation, record sampling and staff interviews.*

## 9. SWOT Analysis

| Strengths | Weaknesses |
|---|---|
| Over 20 years in operation; strong medical expertise; existing EHR assets | Manual intake; long registration time; reported record errors; high nursing administrative effort |

| Opportunities | Threats |
|---|---|
| Mobile and iPad check-in; improved patient experience; efficiency and traceable reporting | Competition from digital urgent-care clinics; privacy/security exposure; unsuccessful technology adoption |

## 10. McKinsey 7S Alignment Assessment

| Element | Current concern / recommended alignment |
|---|---|
| Strategy | Digital intake should directly serve patient experience, information quality and clinical capacity. |
| Structure | Coordinate reception, nursing, clinical, IT and privacy teams with clear decision rights. |
| Systems | Replace manual intake dependencies with controlled EHR-integrated workflows. |
| Shared values | Embed accessible, patient-centered and privacy-conscious |

| | |
|---|---|
| | service in design decisions. |
| Style | Leadership should sponsor phased change and encourage staff feedback. |
| Staff | Plan staffing, training and assisted digital support during rollout. |
| Skills | Build device handling, EHR exception management and data-protection skills. |

## 11. PESTLE Assessment

| Factor | Relevance to Project Pulse |
|---|---|
| Political | Healthcare digital-service priorities and public policy may affect provider planning. |
| Economic | Device, integration, support and training costs require an affordability and value review. |
| Social | Different levels of digital literacy and a preference for convenient check-in require inclusive channels. |
| Technological | EHR compatibility, interoperability, device management and cybersecurity influence feasibility. |
| Legal | The case explicitly specifies HIPAA; applicable jurisdiction and detailed legal obligations require confirmation. |
| Environmental | Potential reduction in paper consumption, balanced against device energy use and electronic waste. |

*External PESTLE implications are contextual considerations, not independently verified facts about CarePlus's location, laws or market.*

## 12. Stakeholder Analysis

| Stakeholder | Interest / responsibility | Engagement |
|---|---|---|
| Patients including older patients | Simple, confidential, accessible registration | Usability tests; assisted pilot sessions |
| Front desk and nurses | Less duplicate typing; clear exception procedures | Workflow workshops; training; pilot feedback |
| Doctors / Chief Medical Officer | Timely, accurate data for care | Validate clinical data availability and escalation |
| IT lead / EHR administrator | Reliable architecture and integration | Technical feasibility and test plan |
| Hospital management | Patient outcomes, operations, cost and risk | Approve scope, KPIs and phased investment |
| Privacy / compliance lead | Legal obligations and access control | Privacy impact assessment; sign-off |
| Technology vendor (if selected) | Specifications and support responsibilities | Solution review, integration and support agreement |

## 13. Proposed Business Requirements (BRD Summary)

| ID | Business requirement | Proposed acceptance measure |
|---|---|---|
| BR-01 | Reduce excessive intake delays | Registration duration measured consistently; target ≤10 minutes after stabilization |
| BR-02 | Improve intake data accuracy | Error rate target below 2%, subject to |

| | | |
|---|---|---|
| | | baseline validation |
| BR-03 | Reduce nonclinical transcription burden | Baseline 40% nursing-shift typing measure reassessed post-pilot |
| BR-04 | Improve inclusion and patient experience | Assisted mode available; patient satisfaction tracked |
| BR-05 | Preserve historical patient records | Existing EHR history remains accessible and reconciled |
| BR-06 | Comply with applicable privacy/security standards | Compliance review and approval before production |

*Performance thresholds are proposed portfolio targets and must be negotiated with sponsors after baseline measurement; they are not promised or achieved outcomes.*

# 14. Functional and Non-Functional Requirements

## 14.1 Functional requirements

| ID | Requirement |
|---|---|
| FR-01 | Patient can initiate intake on mobile or designated hospital iPad. |
| FR-02 | System validates required data fields and displays correction prompts. |
| FR-03 | System checks for potential existing patient matches before EHR update. |
| FR-04 | Authorized integration securely transmits validated intake data to the existing EHR. |
| FR-05 | Staff can review, correct, escalate and resolve submission exceptions. |
| FR-06 | System records check-in status and audit events for authorized users. |
| FR-07 | Staff can view operating metrics with access appropriate to their roles. |

## 14.2 Non-functional requirements

| Category | Proposed requirement / validation approach |
|---|---|
| Security | Encryption in transit and at rest; least-privilege staff access and multifactor authentication where appropriate. |
| Availability | Define a service-level target and tested downtime fallback with hospital operations. |
| Performance | Load-test form submission and integration latency against measured peak traffic. |
| Accessibility | Accessible text, contrast, input controls and assisted user journeys; evaluate WCAG applicability. |
| Interoperability | Document actual EHR interface/API constraints and run patient-matching and reconciliation tests. |
| Privacy | Record retention, notice/consent, audit logging and legal review based on confirmed jurisdiction. |

# 15. Risk Register and Mitigation

| Risk | Relative priority* | Response |
|---|---|---|

| | | |
|---|---|---|
| EHR integration fails or duplicates records | High | Integration discovery, sandbox tests, reconciliation and rollback |
| Unauthorized access or data exposure | High | Privacy assessment, access controls, encryption, testing and audits |
| Some patients cannot self-register | High | Assisted check-in and fallback workflow |
| Staff resistance or insufficient training | Medium | Early involvement, role-based demonstrations and super-users |
| Device/network outage | Medium | Monitoring, local support and business continuity procedure |
| Scope expansion / cost overrun | Medium | Prioritized MVP, formal change review and milestone gates |

**Qualitative priorities are preliminary analyst judgments pending likelihood/impact scoring with stakeholders.*

## 16. Recommended Strategy and Priorities

7. Prioritize a minimum viable digital intake experience: mobile pre-registration plus an accessible, assisted iPad channel, with clear paper or staff-assisted contingency where necessary.
8. Validate secure EHR integration before widespread deployment: patient matching, field mapping, error handling, historical-record preservation and auditability.
9. Introduce operational change and measurement together: redesign the workflow, train front-desk and nursing staff, and track patient waiting times, errors and staff time to establish whether benefits materialize.

## 17. Proposed Implementation Roadmap

| Stage | Indicative duration | Outputs / go-no-go checks |
|---|---|---|
| Discovery and design | Weeks 1–3 | Validated process map; requirements; EHR feasibility; privacy assessment |
| Prototype and integration build | Weeks 4–7 | Configured forms; staff workflow; interface tests; pilot scripts |
| Controlled pilot and training | Weeks 8–9 | One reception area; supervised patients; defect and usability logs |
| Review and phased expansion | Weeks 10–12 | Pilot KPI comparison; sign-offs; staged expansion if thresholds met |
| Continuous improvement | Ongoing | Weekly KPI review, retraining, security checks and enhancement backlog |

*Schedule is illustrative and depends on procurement, EHR vendor cooperation, regulatory review and scope approval.*

## 18. KPI and Benefits Measurement Plan

| KPI | Case baseline | Proposed target / method |
|---|---|---|
| Registration time | Up to 2 hours (maximum reported) | ≤10 minutes target; median, p90 and peak-time distributions |
| Medical record entry errors | 15% reported | <2% target; sampled audited intake records |
| Nursing time spent typing forms | 40% of shift reported | Lower proportion; time study |

| | | |
|---|---|---|
| | | before/after |
| Patient satisfaction | Not provided | Set baseline; target ≥4.5/5 only if measure adopted |
| Digital completion success | Not provided | Track % submitted without staff intervention |
| Integration errors / duplicates | Not provided | Track failures, reversals and reconciliation exceptions |

Measurement should separate pre-arrival form completion time from on-site check-in time, and distinguish registration delay from clinical waiting time. Compare like-for-like periods and patient segments. No verified post-deployment performance or monetary benefit has been supplied.

## 19. Validation and Testing Plan

- Usability: first-time users, older patients and accessibility scenarios complete representative intake flows.
- Data integrity: duplicate identity, conflicting demographics, missing fields and EHR rejection cases.
- Security: user access boundaries, session timeout, lost or unattended iPad and audit event checks.
- Operational resilience: offline devices, slow connection, EHR outage and assisted fallback.
- User acceptance: clinicians confirm availability of accurate patient information; front-desk staff verify exception workflow.

## 20. Assumptions, Constraints and Open Questions

| Assumptions / constraints | Follow-up validation needed |
|---|---|
| An existing EHR must be retained | Which vendor/version? Is a supported integration interface available? |
| HIPAA is cited in the brief | Where is the hospital legally located and which privacy regimes apply? |
| Digital intake devices would be introduced | What budget, procurement rules and device-management controls exist? |
| Reported 15% error rate / 40% typing time are case figures | What sample period, operational definitions and evidence support them? |
| Targets and costs are not approved | What sponsor thresholds, constraints and business case will be accepted? |

## 21. Deliverables and Portfolio Reflection

The portfolio outputs from this exercise are: documented business problem; As-Is and To-Be workflows; PPT gap assessment; Fishbone causes; SWOT, McKinsey 7S and PESTLE analyses; stakeholder needs; proposed BRD summary; solution requirements; risk register; phased roadmap; and KPI/testing approach.

The primary lesson is that digitization does not equal process improvement by itself. A mobile app that leaves manual EHR re-entry intact would not resolve the fundamental bottleneck. The Business Analyst's role is to connect patient and staff needs to a redesigned end-to-end workflow, test the assumptions behind recommendations, and define measurable criteria for stakeholder decisions.

## 22. Conclusion

CarePlus's paper-based intake creates a credible case for change, supported by the reported delay, accuracy and nursing-workload problems. A patient-inclusive digital intake platform linked to the existing EHR could address the core failure points, but its value depends on integration feasibility, training, accessible assisted service, privacy controls and measured outcomes. The next practical decision is to approve a discovery and feasibility phase before committing to full implementation.

## Source and Evidence Notes

Primary source: 10Alytics Business Analysis Class 4, "Gap Analysis Fundamentals in HealthTech / Gap Analysis for Project Pulse (Digital Patient Intake)," facilitator Ifeoma Lilicent Collins, user-provided PDF (12 pages). Baseline figures, organizational description, discovery notes, framework assignment and workflow stages originate from that brief, especially pages 4–11.

Additional context: the analyst previously completed a separate CarePlus performance-analysis exercise examining revenue, costs, margins, consultation hours and departmental performance. No underlying results or numeric financial findings from that earlier exercise are included here; this report does not claim quantified financial impact.

Analyst-developed work: proposed workflow design, possible root causes, stakeholder requirements, KPI thresholds, risks and roadmap are analytical recommendations rather than verified implementation facts.